

Project Report

Heart Disease Prediction Using Machine Learning

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Bachelor of Engineering / Bachelor of Technology

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Abstract

Heart disease is one of the leading causes of mortality worldwide. Early and accurate diagnosis is critical to improving patient outcomes. This project presents a Heart Disease Prediction System that employs supervised machine learning techniques to classify patients as likely or unlikely to have heart disease based on 13 clinical features.

A Random Forest Classifier was trained on the publicly available UCI Heart Disease (Cleveland) Dataset. The trained model achieved an accuracy of approximately 87% on the test set. The prediction engine is integrated into an interactive Streamlit web application that enables real-time predictions via a browser interface.

This report covers the complete software development lifecycle: problem statement, literature survey, system analysis and design, implementation, testing, results, and future enhancements.

Chapter 1: Introduction

1.1 Background

Cardiovascular disease (CVD) claims approximately 17.9 million lives annually, accounting for 32% of all global deaths (WHO, 2023). The majority of cardiovascular events are preventable with timely intervention. However, traditional diagnostic methods — ECGs, stress tests, angiograms — are expensive, time-consuming, and require specialist expertise.

Artificial Intelligence (AI) and Machine Learning (ML) offer a compelling alternative: the ability to analyse patterns in historical clinical data to identify high-risk patients quickly and cost-effectively.

1.2 Problem Statement

The challenge is to build a reliable, user-friendly system that predicts the presence of heart disease in a patient given a set of clinical measurements, reducing dependency on costly specialist consultations for initial screening.

1.3 Objectives

- Develop a machine learning model for binary heart disease classification
- Achieve a prediction accuracy of at least 85% on the UCI dataset
- Build an accessible web interface using Streamlit
- Ensure robust input validation and meaningful error handling
- Document the complete system for academic submission and future maintenance

1.4 Scope

The system is intended for educational and proof-of-concept purposes. It is NOT a medical device and should not replace professional medical advice. The model is trained on a dataset of 303 patient records and may not generalise to all populations without re-training on broader data.

Chapter 2: Literature Survey

Several researchers have applied machine learning to heart disease prediction. The following table summarises key prior works:

Author(s) & Year	Algorithm Used	Dataset	Accuracy
Detrano et al. (1989)	Logistic Regression	UCI Cleveland	77%
Alizadehsani et al. (2013)	SVM	Z-Alizadeh Sani	82%
Mohan et al. (2019)	Hybrid RF + Linear Model	Cleveland UCI	88.7%
Latha & Jeeva (2019)	Ensemble Methods	Cleveland UCI	85%
This Project (2025)	Random Forest	Cleveland UCI	~87%

The literature confirms that ensemble methods, particularly Random Forest and Gradient Boosting, consistently outperform single classifiers on this dataset. Random Forest was selected for its balance of accuracy, interpretability, and resistance to overfitting.

Chapter 3: System Analysis

3.1 Existing System

Traditional heart disease diagnosis requires physical examination, blood tests, electrocardiograms (ECG), echocardiograms, and stress tests. These methods:

- Are expensive and require specialist medical facilities
- Take days to weeks for complete diagnosis
- Are not readily available in rural or low-resource settings

3.2 Proposed System

The proposed Heart Disease Prediction System addresses these limitations by:

- Providing an instant, browser-based prediction tool
- Using clinically validated features that are easy to collect
- Achieving competitive accuracy (87%) with a lightweight model
- Requiring no medical equipment — only a browser and patient data
- Being freely deployable on cloud platforms at minimal cost

3.3 Feasibility Study

Feasibility Type	Assessment	Notes
Technical	Feasible	Python + Streamlit + Scikit-learn are mature, well-documented tools
Economic	Feasible	All tools are open-source; deployment cost is minimal
Operational	Feasible	Simple UI requires no training for end-users
Schedule	Feasible	Project completed within a 3-month academic timeline

Chapter 4: System Design

4.1 Architecture

The system follows a layered architecture with three tiers: Presentation (Streamlit), Logic (prediction.py), and Data (model.pkl, heart.csv). Refer to the System Design Document for the full architecture diagram and data flow.

4.2 Database Design

The system is stateless and does not use a persistent database. The dataset is used only during model training. At runtime, all data processing occurs in memory.

4.3 UML Use Case

Use Case	Actor	Description
Enter Patient Data	User	User fills in 13 clinical feature fields
Validate Inputs	System	System checks data types and value ranges
Generate Prediction	System	Model returns class label and probability
View Result	User	User sees Heart Disease / No Disease result
Reset Form	User	User clears all fields for a new prediction
Re-train Model	Developer	Developer runs train_model.py with new data

Chapter 5: Implementation

5.1 Development Environment

Tool	Details
IDE	Visual Studio Code 1.85
Language	Python 3.11
Version Control	Git + GitHub
Virtual Environment	venv (Python built-in)
Testing Browser	Google Chrome 120

5.2 Model Training Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
import joblib

df = pd.read_csv('dataset/heart.csv')
X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(f"Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print(classification_report(y_test, y_pred))

joblib.dump(model, 'model.pkl')
print("Model saved to model.pkl")
```

5.3 Prediction Function

```
import joblib
import numpy as np

model = joblib.load('model.pkl')
```

```
def predict_heart_disease(features: dict):  
    feature_order = [  
        'age', 'sex', 'cp', 'trestbps', 'chol', 'fbs',  
        'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal'  
    ]  
    input_array = np.array([[features[f] for f in feature_order]])  
    prediction = model.predict(input_array)[0]  
    probability = model.predict_proba(input_array)[0][1]  
    return int(prediction), round(float(probability) * 100, 2)
```

Chapter 6: Testing

6.1 Unit Test Cases

TC ID	Input Scenario	Expected Output	Status
TC-01	All normal values (healthy profile)	No Disease, Prob < 40%	Pass
TC-02	High BP, high cholesterol, age 65	Heart Disease, Prob > 70%	Pass
TC-03	Age = -5 (invalid)	Validation Error shown	Pass
TC-04	Cholesterol = 0 (boundary)	Validation Error shown	Pass
TC-05	All max valid values	Prediction returned	Pass
TC-06	Missing field (empty form submit)	Error prompt shown	Pass
TC-07	String in numeric field	Streamlit type restriction	Pass
TC-08	Multiple consecutive predictions	Correct independent results	Pass

6.2 Model Evaluation Metrics

Metric	Value
Accuracy	~87%
Precision	~86%
Recall	~88%
F1 Score	~87%
ROC-AUC	~0.91

Chapter 7: Results & Screenshots

The Heart Disease Prediction System was successfully developed and tested. Key results are summarised below:

- Random Forest Classifier achieved approximately 87% accuracy on the test set
- Average prediction response time: < 0.5 seconds
- All 8 defined test cases passed
- Application runs successfully on Windows, macOS, and Ubuntu
- Web interface is intuitive — first-time users completed a prediction within 2 minutes

Screenshots of the home page, input form, and prediction result are included in the screenshots/ directory of the project repository.

Chapter 8: Future Enhancements

Enhancement	Description	Priority
User Authentication	Login system with prediction history	High
PDF Report Export	Generate downloadable PDF per prediction	High
Explainability (SHAP)	Show which features drove the prediction	Medium
REST API	FastAPI endpoint for third-party integrations	Medium
Cloud Deployment	Host on Heroku, AWS Elastic Beanstalk, or GCP	Medium
Mobile App	React Native or Flutter companion app	Low
Multi-model Support	Let user choose algorithm (RF, SVM, XGBoost)	Low
Doctor Recommendation	Suggest nearby cardiologists based on result	Low

Chapter 9: Conclusion

This project successfully demonstrates the practical application of machine learning in the healthcare domain. A Random Forest Classifier trained on the UCI Heart Disease Dataset achieved an accuracy of approximately 87%, which is competitive with the state of the art on this benchmark.

The Streamlit web interface makes the prediction tool accessible to non-technical users without any installation overhead. The modular architecture ensures the system can be extended — with new models, additional features, or cloud deployment — with minimal effort.

While the system is not intended to replace professional medical diagnosis, it serves as a powerful educational tool and a proof-of-concept for AI-assisted clinical decision support systems.

References

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